STAFF REVIEW · H446 252 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. MAR.

2. The current instruction that is being decoded or executed.

3. Addresses identifying the memory or device location.

4. Before edge Q stays 0; at rising edge D=1 is sampled so Q becomes 1. It changes on the active edge, not merely when D changes.

I Do — source worked steps

1. Predict first — MAR identifies location 40; the address bus carries that location to memory, accompanied by a read control signal.

2. Explain the answer — The data bus transfers instruction content to MDR, then CIR; sequential PC becomes 41.

We Do — guided CHECK

1. Predict first
The initial fetch advances PC from 81 to 82, as in a sequential instruction.
Explain the answer
Executing the unconditional branch overwrites PC with target address 120. The next fetch occurs there, not at 82.

You Do — source mark schemes

1. [6] Address bus identifies memory/device location [2]; data bus transports instruction/data contents [2]; control bus carries read/write/timing signals [2].

2. [6] PC64→MAR64 [1], instruction memory→MDR [1], MDR→CIR [1], sequential PC→65 [1], decode JMP100 [1], execute branch sets PC=100 [1].

3. [6] Both use fetch/decode/execute [1]; arithmetic instruction may update ACC [1]; interrupt signals an event [1]; processor preserves enough execution state [1], transfers control to an ISR [1] and restores/resumes after service [1].

Exit Ticket — source answer

1. True. The jump target becomes the next fetch address.